

MNIST Classification using PCA, FDA, LDA and QDA

2024450

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1 Introduction

In this assignment we perform classification on a subset of the MNIST dataset. Only the handwritten digits **0**, **1** and **2** are considered.

The objective of this task is to:

- Reduce dimensionality using **Principal Component Analysis (PCA)**
- Perform discriminative projection using **Fisher Discriminant Analysis (FDA)**
- Compare classification performance of **Linear Discriminant Analysis (LDA)** and **Quadratic Discriminant Analysis (QDA)**
- Analyze the impact of PCA on classification accuracy

Each MNIST image has a resolution of 28×28 pixels which corresponds to 784 features. Working in such a high dimensional space is computationally expensive, therefore dimensionality reduction techniques are applied before classification.

2 Dataset Description

The MNIST dataset consists of handwritten digit images ranging from 0 to 9. In this experiment only the digits:

$$\{0, 1, 2\}$$

are used.

For each class:

- 100 samples are randomly selected for training
- 100 samples are randomly selected for testing

Therefore,

$$\text{Total Training Samples} = 300$$

$$\text{Total Test Samples} = 300$$

Each image is converted into a vector of size:

$$784 = 28 \times 28$$

Thus the training data matrix becomes:

$$X \in \mathbb{R}^{784 \times 300}$$

3 Data Preprocessing

The following preprocessing steps were applied:

1. Images were converted to feature vectors by stacking pixels.
2. Pixel values were normalized to the range:

$$[0, 1]$$

using

$$x = \frac{x}{255}$$

3. The dataset was organized into the matrix:

$$X \in \mathbb{R}^{784 \times 300}$$

4 Principal Component Analysis (PCA)

PCA is used to reduce dimensionality while preserving the maximum variance of the data.

4.1 Mean Calculation

The mean vector of the dataset is computed as

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

where $N = 300$.

4.2 Mean Removal

The centered data matrix is obtained as

$$X_c = X - \mu$$

4.3 Covariance Matrix

The covariance matrix is computed as

$$S = \frac{X_c X_c^T}{N - 1}$$

4.4 Eigen Decomposition

The eigenvectors and eigenvalues of the covariance matrix are computed:

$$S u_i = \lambda_i u_i$$

The eigenvectors are sorted in descending order of eigenvalues.

4.5 Dimensionality Reduction

The projection matrix is constructed using the top p eigenvectors:

$$U_p = [u_1, u_2, \dots, u_p]$$

The transformed data is:

$$Y = U_p^T X_c$$

The value of p is chosen such that at least **75% variance** is retained.

5 Reconstruction

The reconstructed samples are obtained by

$$\hat{X} = U_p Y + \mu$$

The reconstruction error is measured using Mean Squared Error (MSE):

$$MSE = \frac{1}{d} \sum_{i=1}^d (x_i - \hat{x}_i)^2$$

Reconstruction results for five samples are shown in output

6 Fisher Discriminant Analysis (FDA)

FDA is used to find a projection that maximizes class separability.

6.1 Within Class Scatter

$$S_W = \sum_c \sum_{x_i \in C_c} (x_i - \mu_c)(x_i - \mu_c)^T$$

6.2 Between Class Scatter

$$S_B = \sum_c N_c (\mu_c - \mu)(\mu_c - \mu)^T$$

where

- μ_c is the mean of class c
- N_c is the number of samples in class c

6.3 Optimal Projection

FDA finds projection matrix W that maximizes:

$$W = \arg \max \frac{\det(W^T S_B W)}{\det(W^T S_W W)}$$

This leads to the generalized eigenvalue problem:

$$S_W^{-1} S_B w = \lambda w$$

The top eigenvectors form the projection matrix.

7 Classification

Two classifiers were evaluated.

7.1 Linear Discriminant Analysis (LDA)

The LDA discriminant function is:

$$g_i(x) = x^T \Sigma^{-1} \mu_i - \frac{1}{2} \mu_i^T \Sigma^{-1} \mu_i$$

where

- μ_i is the class mean
- Σ is the shared covariance matrix

The predicted class is

$$\hat{y} = \arg \max_i g_i(x)$$

7.2 Quadratic Discriminant Analysis (QDA)

QDA uses class-specific covariance matrices.

The discriminant function is:

$$g_i(x) = -\frac{1}{2}(x - \mu_i)^T \Sigma_i^{-1}(x - \mu_i) - \frac{1}{2} \log |\Sigma_i|$$

8 Accuracy Metric

Accuracy is defined as:

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

Mathematically,

$$Accuracy = \frac{1}{N} \sum_{i=1}^N I(\hat{y}_i = y_i)$$

where I is the indicator function.

9 Effect of PCA Variance

Experiments were conducted with:

- 75% variance
- 90% variance
- first two principal components

Higher variance generally retains more information, leading to improved classification performance. However, increasing dimensionality also increases computational cost.

10 Conclusion

In this assignment we implemented PCA and FDA for dimensionality reduction and compared LDA and QDA classifiers.

Key observations:

- PCA successfully reduced the dimensionality of the dataset while retaining most of the variance.

- FDA provided better class separation because it explicitly maximizes between-class variance while minimizing within-class variance.
- LDA performed well when the covariance matrices of classes were similar.
- QDA was more flexible but slightly more sensitive to noise due to class-specific covariance estimation.

Overall, combining PCA with discriminant analysis provides an effective pipeline for classification of high-dimensional image data.